

7: Classification metrics:

(A) TP - 45 TN - 165

FP - 15 FN - 25

(B) Accuracy: $\frac{45 + 165}{45 + 165 + 15 + 25} = \frac{210}{250} = \underline{\underline{0.84}}$

Precision: $\frac{45}{45 + 15} = \frac{45}{60} = \underline{\underline{0.75}}$

Recall: $\frac{45}{45 + 25} = \frac{45}{70} = 0.642$

$F_1 = 2 * \frac{0.75 * 0.642}{0.75 + 0.642} = 2 * \frac{0.481}{1.392} = 2 * 0.345$

$F_1 = 0.691$

Specificity (just in case it's needed and not mentioned in the Q by accident)

$\frac{165}{165 + 15} = \underline{\underline{0.916}}$